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EDUCATION
Harvey Mudd College, Claremont, California
Bachelor of Science in Engineering with Distinction May 2026

RELEVANT COURSEWORK
Mechanical Design, Materials Engr., Manufacturing Planning/Execution, Mechanics & Wave Motion, Multivariable Calculus, Data / Probability & Stats for Engr., Differential Equations, Fluid Mechanics, Electrical & Magnetic Circuits / Devices, Thermodynamics, Intro to Engr. Design & Manufacturing, Advanced Systems Engr., Continuum Mechanics, Computer Science

SKILLS
CAD/CAE Software: SolidWorks, AutoCAD, Fusion, Onshape, COMSOL Multiphysics
Programming & Data Analysis: MATLAB, Python, LabView, SystemVerilog, Javascript, C, Arduino (C++), SSH
Advanced 3D printing (SLA, SLS), CNC machining (ShopBot, mill, lathe), laser cutting, welding, soldering, and comprehensive manual shop equipment (saws, grinders)
Languages: Spanish (Native) and French (Limited Proficiency)

EXPERIENCE
Blue Origin, Claremont, California August 2025 - May 2026
Mechanical Team Lead - Automated ground-side umbilical mating procedure for New Glenn rocket
- Directed the mechanical sub-team in engineering an automated actuation system, utilizing a 4-degree-of-freedom (4-DOF) kinematic architecture to achieve millimeter-level alignment of umbilical panels during dynamic launch scenarios.
- Evaluated competing design concepts through comprehensive mechanism trade studies, leveraging CAD and Finite Element Analysis (FEA) to optimize structural integrity and validate load-bearing capabilities.
- Synchronized mechanical hardware interfaces with the master control algorithm, collaborating closely with controls and sensor subsystems to define the overarching system architecture.
- Translated theoretical kinematic models into physical testbeds by rapidly prototyping scaled sub-assemblies, establishing a continuous feedback loop to refine the final hardware installation plan.

Flow Imaging Lab Research, Claremont, California August 2024 - May 2026
CAD Lead & Trajectory Designer - Research on archerfish-inspired propulsion system for aerial-aquatic robots to replace chemical-based systems with a sustainable, eco-conscious data sampling approach
- Designed and fabricated improvement to existing experimental apparatus and mounting fixtures using SolidWorks, serving as lab's CAD Manager.
- Quantified hydrodynamic forces and torques during water-exit events utilizing a 6-axis load cell.
- Analyze fluid dynamics in Matlab by implementing Particle Image Velocimetry to visualize flow patterns and utilizing Computational Fluid Dynamics to validate experimental data and inform trajectory design.
- Maintain and optimize data acquisition equipment and experimental tank to assure operational readiness and data integrity.

Harvey Mudd Automotive Club (MACH), Claremont, California September 2022 - May 2026
Founding Member & Mechanical Lead, Custom Automotive Build
- Co-founded the college's first automotive club to demonstrate values of sustainability and resourcefulness. Conceptualized an eco-friendly vehicle platform originally designed to collect and repurpose campus scrap materials.
- Led the mechanical integration of a hybrid powertrain. Collaborated with teammates to power the primary electric motor, while also coupling an internal combustion engine to the rear axle.
- Architected steering and braking subsystems. Utilized MIG welding and makerspace equipment to fabricate frame from steel tubing and angle iron.

Indomo TX, Claremont, California January 2025 - May 2025
Mechanical Engineer - Redesigned a next-generation autoinjector for a clinical-stage company's novel, at-home acne treatment
- Engineered mechanical improvements, defined system assembly plans, and optimized the device for Design for Manufacturability.
- Implemented a critical safety interlock integrating user activation mechanism with a skin-contact sensor, eliminating risk of accidental device discharge.
- Developed a decoupled mechanical system to isolate needle deployment from drug delivery, calibrating the mechanism's timing to elevate user safety and guarantee reliable, repeatable operation.
- Accelerated iterative development cycles by rapidly 3D prototyping subsystem CAD models, conducting physical testing to verify mechanical tolerances and ensure alignment with clinical usability requirements.